# Project-X: Forcing Visual Grounding

Causal-Aware Few-Shot Learning in Vision-Language Models
via Counterfactual Retrieval and Attention Modulation

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# Outline



**Workflow**

Causal retrieval $\rightarrow$ $k$-shot OpenFlamingo $\rightarrow$ LoRA grounding $\rightarrow$ measure accuracy + causal image use

# Problem Statement

**VLM In-Context Learning is fragile in three ways:**

1. Modality imbalance

Text-driven ICL; images largely ignored (weak cross-attention).

2. Spurious retrieval

RICES $=$ passive similarity $\Rightarrow$ *non-causal* demos.

3. Many-shot paradox

More shots $\Rightarrow$ worse; optimum $=$ narrow few-shot window.

**Goal:** force *causal visual grounding*.

# State of the Art

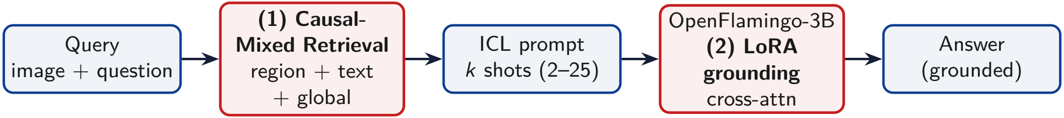
**Prior work**

- Flamingo / OpenFlamingo — gated cross-attn ICL
- RICES — CLIP-similarity retrieval
- ICL analyses — visual modality under-used
- Counterfactual CIR — counterfactuals help
- Many-shot studies — more can hurt

**Gap addressed**

- Prior: retrieval *or* architecture
- Ours: causal retrieval + grounding loss
- Validated by *causal reliance*, not only accuracy

# Proposed Method — Dual Novelty




## (1) Causal-Mixed Retrieval

- Tri-weighted: region+text+global
- Class-diversity filter
- Bounded shot window

## (2) Grounding Penalty

- Counterfactual grounding loss
- LoRA on gated cross-attn
- Uses the *correct* image

# Method (1) — Causal-Mixed Retrieval

Score each demo vs. query — region + text + global:

$$\mathrm{score}(c) = 0.4\ \cos\left(f_q^{\mathrm{loc}}, f_c^{\mathrm{loc}}\right) + 0.3\ \cos\left(f_q^{\mathrm{txt}}, f_c^{\mathrm{txt}}\right) + 0.3\ \cos\left(f_q^{\mathrm{glob}}, f_c^{\mathrm{glob}}\right)$$

- CLIP ViT-B/32 features; region = bbox crop (else global)
- Class-diversity filter, then back-fill
- Shots clamped $[2, 32]$; tested $k \in \{2, 5, 15, 25\}$

**Why it helps**

Region ⇒ local causal match; diversity ⇒ no near-duplicates.

# Method (2) — Cross-Modal Attention-Forcing Penalty

LoRA on gated cross-attn; $\ell_b = \mathrm{NLL}(\text{real})$, $\ell_c = \mathrm{NLL}(\text{wrong})$:

$$\mathcal{L} = \underbrace{\ell_b}_{\text{task}} + \lambda_g \underbrace{\mathrm{ReLU}(m - (\ell_c - \ell_b))}_{\text{wrong img worse by } m} + \lambda_e \underbrace{H(\text{attn}_{\text{answer}})}_{\text{sharpen attention}}$$

- Grounding: wrong image must be worse by margin $m$
- Entropy: sharpen attention on answer tokens
- PEFT only: **3.24M / 2.69B** $\approx$ **0.12%** params

# Dataset

**Train — grounding adapter**

**VQAv2** (val stream, $N = 2000$) — general VQA, no benchmark leakage

**Zero-/few-shot evaluation** (held-out):

**GQA — testdev balanced**

- Compositional, low language bias
- Tests grounding (spatial)
- `lmms-lab/GQA`

**AwA2 — Animals w/ Attributes**

- 50 classes, 85 attributes
- Tests causal retrieval
- `KRadim/AwA-Pose-Lite`

$n = 150$ queries / ablation / shot level

# Experimental Setup

## Model & training

- OpenFlamingo-3B: CLIP ViT-L/14 + MPT-1B, cross-attn every layer.
- LoRA: $r$=16, $\alpha$=32, dropout 0.05 on `to_q/to_kv/to_out`.
- AdamW, lr $2\times10^{-5}$, wd 0.01, 1 epoch, 150 steps.
- $\lambda_g$=0.5, margin $m$=0.5, $\lambda_e$=0.05.
- Seed 42; eval in fp16; CLIP ViT-B/32 retriever.

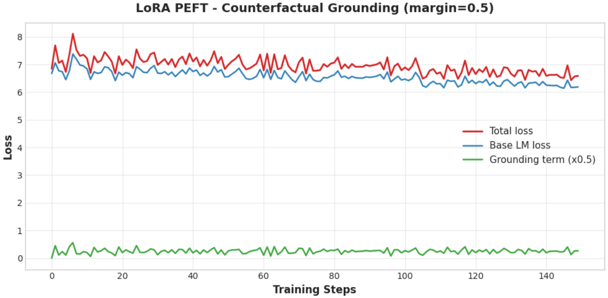
## 2×2 factorial ablation

| | adapter **OFF** | adapter **ON** |
|---|---|---|
| **random** retr. | Baseline | Ablation 3 |
| **causal** retr. | Ablation 1 | Ablation 2 |

**Ab1** = retrieval; **Ab3** = grounding; **Ab2** = full.

**Metrics:** EM (= GQA accuracy), F1, answer confidence, reliance $\Delta$NLL = NLL(blind)−NLL(real).

# Training — Counterfactual Grounding Curve

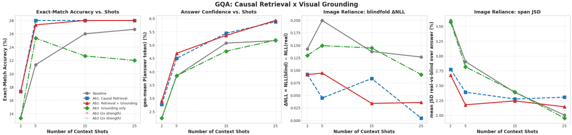



- Base LM loss **6.68** → **6.19**
- Grounding term active (mean ≈ 0.45)
- Attn entropy **3.48** → **2.68** (sharper)
- Avg total loss ≈ 6.96

⇒ uses correct image, no LM damage

# Model Evaluation — GQA (Exact-Match %)

| Config | 2 | 5 | 15 | 25 |
|---|---|---|---|---|
| Baseline | 13.3 | 21.3 | 26.0 | 26.7 |
| Ab1 Causal retr. | **17.3** | **28.0** | **28.0** | **28.0** |
| Ab2 Retr.+Ground | **17.3** | 27.3 | **28.0** | **28.0** |
| Ab3 Ground only | 13.3 | 25.3 | 22.7 | 22.0 |

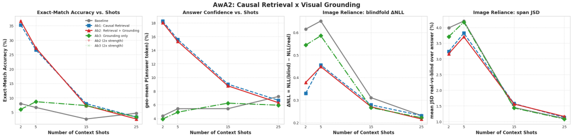



- Causal retrieval: **+4 to +7** EM, every *k*
- Grounding alone (Ab3) lags
- Gains saturate by 5–15 shots

# Model Evaluation — AwA2 (Exact-Match %)

| Config | 2 | 5 | 15 | 25 |
|---|---|---|---|---|
| Baseline | 8.0 | 6.7 | 2.7 | 4.7 |
| Ab1 Causal retr. | **35.3** | **26.7** | **8.0** | 3.3 |
| Ab2 Retr.+Ground | **36.7** | **27.3** | 7.3 | 2.7 |
| Ab3 Ground only | 6.0 | 8.7 | 7.3 | 3.3 |

- 2-shot: 8.0 → **36.7** (+28.7)
- Confidence 4.4% → 18%
- **Many-shot collapse** beyond 5 shots



# Grounding Diagnostics — Does the model use the image?

**Blindfold (5-shot):** swap query image. Gap = EM(real)−EM(mismatch); bigger = more image use.

**AwA2**

| Config | real | mism. | **gap** |
|---|---|---|---|
| Baseline | 8.3 | 0.0 | +8.3 |
| Ab1 | 26.7 | 8.3 | +18.3 |
| Ab2 | 30.0 | 5.0 | **+25.0** |
| Ab3 | 6.7 | 0.0 | +6.7 |

**GQA**

| Config | real | mism. | **gap** |
|---|---|---|---|
| Baseline | 18.3 | 13.3 | +5.0 |
| Ab1 | 28.3 | 26.7 | +1.7 |
| Ab2 | 25.0 | 23.3 | +1.7 |
| Ab3 | 18.3 | 13.3 | +5.0 |

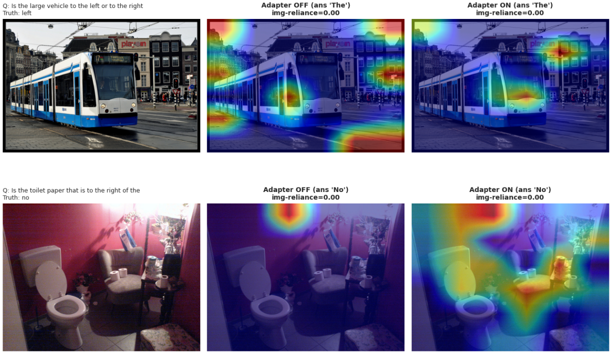
- AwA2: Ab2 gap **+8.3** → **+25.0** (triples)
- GQA: smaller gap — strong language priors

# Grounding Diagnostics — Classification & Saliency

**AwA2 likelihood classification**
(2-shot, $n = 20$, rank by log-lik.)

| Config | top-1 | top-5 |
|---|---|---|
| Baseline | 25.0 | 45.0 |
| Ab1 | **45.0** | **65.0** |
| Ab2 | **45.0** | **65.0** |
| Ab3 | 10.0 | 50.0 |

- Top-1 nearly **doubles** (25 → 45%)
- Saliency → queried object (adapter ON)



# Conclusions & Future Work

## What worked

- Retrieval = main driver: **+28.7** (AwA2), **+6.7** (GQA)
- Grounding $\Rightarrow$ causal: blindfold gap **+8.3** $\rightarrow$ **+25.0**
- Many-shot paradox reproduced

## Limitations & future work

- Grounding adds reliance, not much accuracy (Ab2 $\approx$ Ab1)
- Add GQA Validity / Plausibility / consistency
- Scale $n$, larger backbone, learned retrieval weights

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